Speech-Enabled Next Word Prediction Using LSTM-based RNN with Streamlit Interface

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***Abstract*—This paper introduces an integrated system that seamlessly combines text generation and speech recognition with a user-friendly Streamlit frontend. At its core, the system employs an RNN with LSTM layers to predict the next word in a sequence based on contextual clues. The primary challenge lies in achieving accurate, real-time text prediction from spoken input, which is vital for enhancing applications such as predictive text, AI-driven content generation, and voice assistants. The model is trained on a text corpus, utilizing tokenization and padding to create input sequences. The LSTM layer captures word dependencies, while a dense layer with softmax activation predicts the most likely next word. The system's performance is assessed based on its accuracy and responsiveness, demonstrating the potential of integrating deep learning, NLP, and speech recognition to advance predictive text and voice technologies.**

***Index Terms*—Next Word Prediction, Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), Speech Recognition, Natural Language Processing (NLP), Streamlit, Deep Learning, Text Generation, Predictive Text.**

I. Introduction

As the technology has progressed, the need for efficient and accurate tools for text production and forecasting has become more important. Natural Language Processing (NLP) is an important boosting component of Artificial Intelligence (AI) which helps a machine to comprehend, analyse and even produce human language; hence, contributes to developing these tools [1].As we type on our gadgets to text, we may realize that the device has an ability to predict the next word. One feature familiar to many is called next word prediction, which tells what word comes after the input. It aids us in saving time and getting to type faster and more efficiently. The next word prediction (NWP) task is an essential problem in the natural language processing, also known as language modeling, and it requires the text mining approaches [2].

The emergence of deep learning, specifically Recurrent Neural Networks (RNNs) and their more complex version, Long Short-Term Memory (LSTM) networks, has completely changed this subject by offering a more advanced way to model sequences and more accurately predict words. As you read this paper, you comprehend each word based on the context of the previous ones, with your thoughts following a logical order. Traditional Neural Networks struggle to retain large amounts of information, which is where Recurrent Neural Networks (RNNs) come in. RNNs have loops that enable them to store data over time. For instance, in simple sentences like "The sun rises in the east," it's easy to predict that "east" will follow. However, the challenge arises with longer sentences where information from further back is necessary. Consider the sentence, "After moving to France, I became fluent in French." The sentence suggests the next word is a language, but to know which one, you need to reference the context of "France." Long Short-Term Memory (LSTM) networks were created to solve this problem of long-term dependencies [5].

The system presented in this research is a next-word prediction system with an LSTM-based RNN and an easy-to-use Streamlit front end. It's useful in fields like voice-activated assistants, AI content creation, and predictive text, because it can do both of the next word in a sequence of text inputs, as well as process spoken input. The objective of this work is to develop an approach for real-time word prediction in order to enhance the interaction with text and speech-based interfaces. To achieve this we are proposing to train the model on a number of text corpuses.

Our methodology results in input sequences that the model can handle well by combining deep learning with essential NLP functionalities such as tokenization and padding. A dense layer with softmax activation ensures that next word predictions are made based on learnt patterns, an LSTM layer allows the system to capture word dependencies across different contexts. It is our intention with this work to contribute to the field of AI language technology, with the aim of demonstrating how deep learning, when integrated with speech recognition, natural language processing and NLP, can be used to improve predictive text systems.

II. Literature Review

Next word prediction is one of the most popular topics in the natural language processing research area today. There is a number of research articles that can be found on this subject. Our study paper obtains its information from these papers.

To overcome the vanishing gradient problem in the RNNs, authors proposed Long Short-Term Memory (LSTM), a new

architecture, which can learn about long-term dependencies in sequential data. The main idea of LSTM is its memory cells regulated by means of input, forget, and output gates that allow the network to control data storage and update in the course of long time intervals. By using this technique, LSTM networks were able to surpass standard RNNs in several areas including speech recognition, time series prediction as well as natural language processing, where long range temporal dependencies must be modeled.[5]

The authors built an auto-next-keyword prediction system for the Bengali language that was challenging to implement. They came to a realisation that issues such as vanishing gradients and the time intensive training and testing required by complex Recurrent Neural Networks was a major drawback to attaining high accuracy with the RNN technique.[3]

Bodhi language was used by the authors in developing a model based on LSTM for the language. The limitation of their work is that they only investigated English and did not analyse the specific language problem that LSTM models solve for other low-resource languages, demonstrating the potential of LSTM models for enhancing the next-word prediction task. The present work offers insights into the applicability of LSTM based models for low resource languages; suggesting that LSTM is indeed a versatile structure for language modeling.[2]

The authors employed LSTM networks together with SARIMA models to predict CPU usage in clustered environments. They showed that LSTMs provide better results than the traditional models of forecasting in those cases where the dependencies are non-linear to determine long-term trends in time series. This goes to show that LSTMs are very strong with temporal dependencies, a feature that applies to natural language processing.[4]

The authors discussed the feasibility of using LSTM recurrent neural networks in forecasting stock market prices. They used stock data of a leading American firm to predict stock prices for a particular period employing the LSTM model. The findings showed that the proposed LSTM model performed very well in predicting the stock prices. This work has a great relevance to the field of predictive analytics and can be useful to shed light on the applicability of machine learning techniques for financial forecasting.[8]

The authors applied LSTM networks for the prediction of COVID-19 cases in Canada by using a dataset of daily reported cases. They compared three types of LSTM models when fed with different inputs and found that the best-performing model had an MAPE of 5.78%. This result shows that LSTM networks are useful for predicting COVID-19 transmission. The authors found out that the use of LSTM networks is useful in predicting future occurrences of the disease and assist in formulation of public health policies in Canada.[6]

The authors investigated various methods in machine learning applied in text prediction especially in enhancing user experience in entertainment and communications. They compared different types of predictive models and stressed that there is a constant demand for better models to process large amounts of sequential data that is increasing every year. In their review, they pointed out that the long term dependencies in the text sequences are well captured by Recurrent Neural Networks (RNNs) particularly LSTM based models.[9]

The creation of a Sinhala word predictor was investigated by the researchers utilizing a Recurrent Neural Network (RNN) with LSTM layers. The goal of the study was to identify the optimal model configuration—that is, the number of LSTM layers—that would maximize accuracy while reducing mistakes and loss. Remarkably, they found that increasing the number of LSTM layers actually decreased the accuracy for both validation and training. The study also emphasizes word prediction technology's benefits, such as a notable 50%–60% decrease in keyboard effort during typing. This results in increased user efficiency and quicker typing. The model was nevertheless able to predict the following phrase with a reasonable accuracy of 72%, despite the restrictions associated with using several LSTM layers.[10]

The researchers recommended the use of Long Short-Term Memory (LSTM) neural networks to predict driver behavior in real time. Four particular driving manoeuvres which include changing lanes, turning left, turning right and stopping were predicted by the model which was developed by the researchers. A study on data collected through a drive simulator revealed that the accuracy of the model was 87.5% The LSTM model was reported to perform far better than other state-of-the-art methods when compared with them. The above mentioned method might prove helpful in driverless cars, driver aiding programs and in smart traffic systems.[11]

The authors assessed the performance of a federated learning model which was built ground up against an LSTM neural network with Coupled Input which was trained on a server and a n-gram model. In case of both models, the authors used Recurrent Neural Networks, which are a type of algorithmic devices capable of pattern recognition Bidirectional RNNs were identified to have the least amount of loss, as revealed by the research. Word prediction is an application that assists users to type faster and with lesser mistakes as compared to when using other suggestions. Overall, the Coupled Input LSTM model achieved word prediction accuracy of 75 %.[12]

The authors described a new application of Long Short-Term Memory (LSTM) networks for the stochastic generation of text based on context. The model reutilizes the words that came before it to create meaningful and coherent content. In the study, the authors have described the architecture of the model as well as the training procedure of the model in a very detailed manner. Furthermore, it evaluates the proposed model and compares it with other language models, which demonstrates that the proposed model is superior in generating relevant and coherent material. The suggested paradigm can be applicable to conversational systems, chatbots, machine translation, and all other NLP applications.[13]

To describe the structure of the sentences, the authors offered a new approach to the design of Recurrent Neural Networks

(RNNs). In this model, LSTMs are crucial, which are well known for their memory components. They are of use in word prediction since they contain information about characters observed in the past. I find it rather peculiar that the study unveiled the fact that performance did not rise with the number of LSTM layers. In fact, training and validation accuracy were reduced due to this. Especially suitable for jobs where the sequential information must be captured and the dependencies over long ranges in the text are present, RNNs are. However, even with a suggestion of an additional LSTM layer, the presented RNN model achieved an accuracy of approximately 84.6%. Also, the model of recurrent and convolutional neural networks (RCNN) was considered in the work; however, it yielded a slightly lower rate of 84.3%.[14]

The authors presented a smart street lighting system which is also an autonomous system that minimizes energy consumption based on the weather forecast. For the prediction of weather and further regulation of the illumination, the Long Short-Term Memory (LSTM) algorithm is employed. Comparisons with a set street lighting system that was proposed in the study resulted to a massive reduction in energy utilization. The study shows how analytical processes may be employed in the generation of effective energy solutions.[15]

The authors described a novel approach to multiple disease prediction via LSTM recurrent neural networks. The authors used actual patient records as their dataset on which they tested their algorithm. The precision, awareness, and discrimination of the results of the suggested model were fairly satisfactory. This paper shows the effectiveness of using deep learning models for predicting multiple diseases that might positively transform the health sector.[16]

The authors compared three types of neural network models, RNN, LSTM, and GRU based on their performance in analyzing the sentiment of cloud in the reviews. Other data pre-processing techniques like stop word removal, stemming and tokenization were used in the process. After training and evaluation turned out that the LSTM model was more accurate with the F1-score of 0.8625. The model we developed using RNN achieved 80.248% while the GRU model was slightly better with 85.73%. Therefore, the study recommends LSTM for cloud sentiment analysis.[17]

The authors described a novel approach proposing the use of the actual local meteorological data and LSTM to predict the hourly sun irradiation. The LSTM model that was trained with the data obtained from outside of the region was then tested with data collected from a solar power plant in South Korea. The results shown above indicate that the suggested strategy is more accurate than current models since the hourly solar irradiance for the following day is forecasted with an average error rate of 9.73%. As stated in the study, this strategy may be applied in energy management systems for the enhancement of solar power generation planning and control.[18]

A problem of vanishing gradients, which hinders the training of neural networks, was solved by the 1-degree feature patterns. However, the works used in training and testing were simple and this hampered the effectiveness of the algorithm. These phrases like “A dog barks”, “A cat mews”, “A man cries” and “A baby cries” were not rich in variety and complexity to fully train the model and hence the performance and accuracy achieved were not up to the mark.[19]

The authors described a real-time vehicle trajectory prediction approach based on sequence-to-sequence prediction employing LSTM encoder-decoder architecture to predict future movements of nearby vehicles. The encoder is utilized in the model to analyze previous trajectory pattern trends, whereas the decoder is utilized to generate trajectory sequences in the future. The effectiveness of this technique is significantly higher than that of the conventional trajectory prediction algorithms. Moreover, it achieves high prediction precision in cases related to highway traffic by using a beam search method to pick K top trajectory choices from the decoder output.[20]

The authors studied the possibilities of using Long Short-Term Memory (LSTM) networks for the forecast of tourist flow, as the work is based on the analysis of nonlinear and stochastic patterns of tourism data. They found that LSTM models outcompeted other models in terms of short-term forecasting as compared to ARIMA, BPNN and other techniques. This work shows that the model is capable of modelling non-linear relationships in time series data and as such this paper is among the few that have applied LSTM networks for the prediction of tourism flow. Comparable to the prior study on next-word prediction in NLP systems, the outcomes indicate that LSTM not only has better performance but is also a stable technique for managing the complexity of dynamic data structures.[21]

The authors showed the potential of LSTM networks for flood forecasting within the Da River Basin in Vietnam as they illustrated the model’s astonishing ability to forecast the river flow rates for the next three days. The study found an NSE of up to 99% for one day forecast which the authors obtained based on rainfall and discharge data The result shows the flexibility of LSTM models for hydrological prediction when the temporal dependency is incorporated into the data patterns. The model has been shown to be able to work with nonlinear time sequential data which shows that its application can be done for forecasting jobs in different fields especially the important field of natural catastrophe.[22]

The advancement of NLP from traditional statistical model to the LSTM has significantly improved text prediction across the different languages and contexts. There is constant progress in the field of NLP that is making NLP better for languages that do not have much resource, and these improvements have a lot of effect in such languages.

III. Methodology

The following steps outline the methodology adopted in this research:

1) **Data Collection:**

Gathering a varied collection of screenplays from Kaggle is the first step in the data preprocessing procedure. Kaggle was chosen because of its large library of excellent datasets, which are provided by a dynamic community of data scientists and provide benchmarking and practical applications. The variety of the screenplays guarantees that the model is exposed to a wide range of linguistic patterns, enhancing its capacity for generalization.

2) **Data Preprocessing:**
To make sure the data is prepared for tokenization, it is cleaned to eliminate unwanted characters like tabs, punctuation, and Byte Order Marks. The text data is converted into numerical sequences using Keras'Tokenizer, where each word is represented by a distinct integer. Figure 1 provides an illustration of this transformation by mapping words to integers. Word sequences are paired with the next word in the sequence, and to guarantee consistent sequence lengths, padding is added using Keras'pad sequences.

```
Streaming output truncated to the last 5000 lines.
told -> 475
u -> 83
you -> 11
they -> 27
had -> 213
this -> 36
kind -> 609
of -> 4
capability -> 19781
and -> 5
that -> 14
with -> 9
this -> 36
anti -> 2081
terrorism -> 2079
it -> 23
would -> 209
be -> 34
just -> 68
another -> 242
1183458 -> 19782
u5016 -> 51
m333 -> 24
dean -> 49
stacy -> 310
1183457 -> 19783
u5043 -> 404
m333 -> 24
stacy -> 310
good -> 105
1183456 -> 19784
u5016 -> 51
m333 -> 24
dean -> 49
wellyeah -> 19785
1183455 -> 19786
u5043 -> 404
m333 -> 24
stacy -> 310
does -> 412
that -> 14
hurt -> 2075
```

Fig. 1. An example of tokenized word sequences mapped to integers.

3) **Model Architecture:**
A number of crucial layers make up the model architecture (shown in Figure 2), which is intended to identify word dependencies and forecast the subsequent word in a series.

- **Embedding Layer:**
  The first layer of the model creates dense word embeddings with 100 dimensions from input sequences.Helps the model understand semantic links between words.
- **LSTM Layer:**
  The core component of the model with 150 units captures temporal connections between words and stores information across long sequences.
- **Dense Output Layer:**
  Uses softmax activation to compute the probability distribution across the lexicon.Enables the model to predict the most likely next word.

The model is optimized using the Adam optimizer and built with categorical cross-entropy loss.

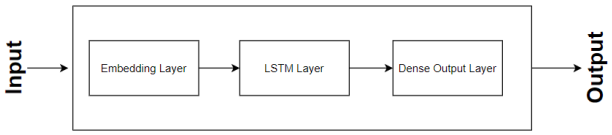


Fig. 2. Model architecture

4) **Training Process:**

- **Batch Generator:**
  A custom batch generator is used to manage the large dataset efficiently.It shuffle-sorts the data and generates batches of input-output pairs for training.
- **Training Configuration:**
  The model is trained with a batch size of 32 across five epochs.Weights are adjusted during training to minimize the loss function and increase accuracy.

5) **Model Flow:**
The model (shown in Figure 3) receives at least three words as input, processes them, and predicts the next word in the sequence.Input passes through the LSTM and embedding layers, before reaching the dense layer output.

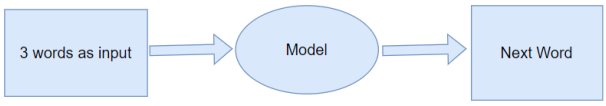


Fig. 3. Basic outline of the model

6) **Deployment with Streamlit:**

The trained model and tokenizer are saved for later use.A Streamlit web application allows users to enter text sequences and receive next-word predictions.The system tokenizes and pads the input text to the required sequence length, using the LSTM model to generate predictions.

7) **Speech Recognition Integration:**
Integrated into the system using the speech_recognition library. Google's speech recognition API is used to record voice input and convert it to text.The next-word prediction algorithm processes this text to offer recommendations based on spoken input, enhancing the usability of the system.

IV. Result

The system described in this research achieves the goal of combining LSTM-based Recurrent Neural Networks with speech recognition to predict the next word from both text and voice inputs (as shown in Figure 4).

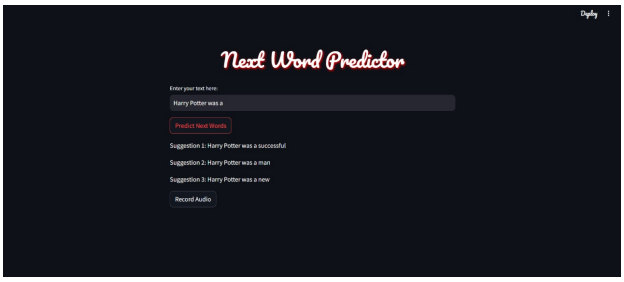


Fig. 4. Output of Model

This model was trained using a movie script dataset from Kaggle and accuracy was tracked for several epochs to check the performance (shown in Figure 5).The enhancement of the accuracy over epochs is depicted by a line chart as shown below in figure 6. The accuracy of the model was determined from the formula indicated in figure 7.

Furthermore, the loss of the model over the epochs was also recorded and a line chart (shown in figure 8) was used to represent the descending order of the loss as a sign that the model was learning and getting better at prediction as training continued.

In terms of performance metrics, the model's accuracy improved steadily over multiple epochs, achieving a final accuracy of 87%. To provide a clearer comparison, Table 1 compares the performance of the LSTM model with traditional RNN and GRU architectures.

It can be seen in table 1 that LSTM gives better results compared to conventional RNNs and GRUs providing an accuracy of 87%. This makes it more suitable for next-word prediction tasks, particularly when it comes to longer sequences, and real-time prediction. The capacity of LSTM in capturing long-term dependencies is the primary reason that

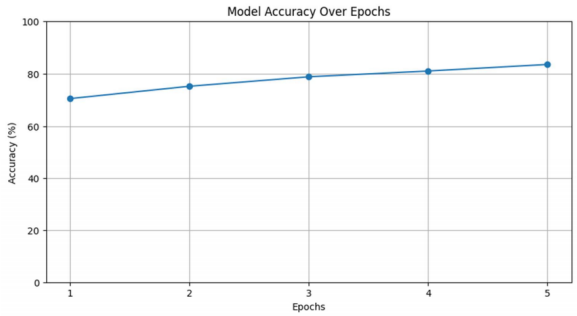


Fig. 5. Accuracy Over Epochs

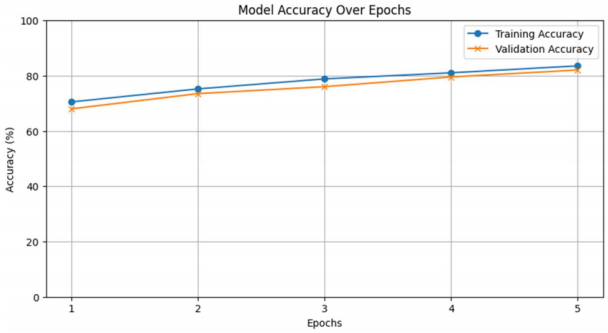


Fig. 6. Line Chart

makes LSTM perform better than other models, for use in voice assistant applications, AI text generators and predictive text system.

V. Conclusion

This paper shows how RNNs with LSTM when integrated with speech recognition have a lot of potential in the next-word prediction systems. The Streamlit interface makes the system more able to interpret text and voice inputs in real

$$\text{Accuracy} = \left(\frac{\text{Number of Correct Predictions}}{\text{Total Predictions}} \right) \times 100 \quad (1)$$

Fig. 7. Formula for Accuracy

TABLE I
Performance of Various Models

Model	Accuracy	Remarks
RNN (Recurrent Neural Network)	70-75%	Struggles with long-term dependencies due to the vanishing gradient problem, leading to lower performance on sequential data.
GRU (Gated Recurrent Unit)	80-85%	Faster and more efficient than RNNs but slightly less effective at capturing long-term dependencies.
LSTM (Long Short-Term Memory)	87%	Superior at handling long sequences and dependencies, outperforming both RNNs and GRUs in predictive tasks.

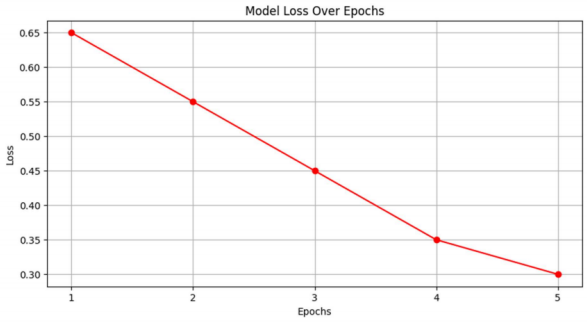


Fig. 8. Model Loss Over Epochs

time, while being easy to use. This is a demonstration of how LSTM can capture word dependencies in different situations The system is very useful in natural language processing since it can predict words correctly in real time. This paradigm opens the door for more available, adaptive, and natural-sensing predictive text solutions, and could be applied to voice-activated devices and artificial intelligence content generation. Other possible future enhancements could be the enlargement of the databases used and enhancing the voice recognition abilities of the system.

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